



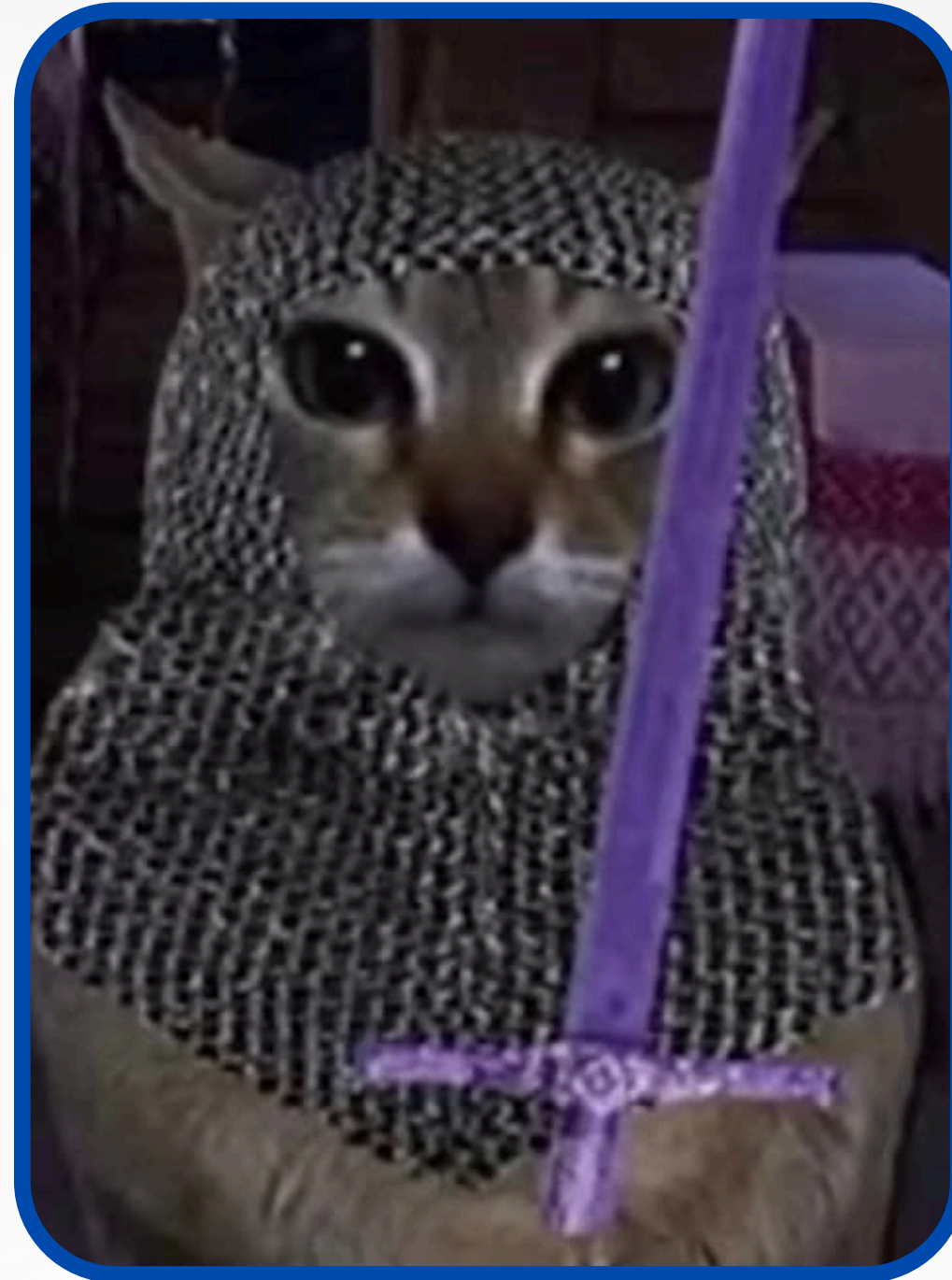
MINI-PROJECT **PRESENTATION**

ABOUT MY PROGRAM

two-player, turn-based combat game (Warrior / Mage / Rogue, inspired by classic RPG duels) played by two clients connecting to a referee server over the network.

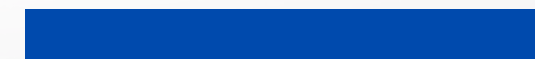
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Each client picks a class, then the two players take alternating turns choosing an action (attack, spell, block, buff, etc.). The server is the single source of truth for game state — it validates every action, applies damage/status-effect rules, and reports the result back to both players with an explicit status code, the same way an HTTP server reports success/failure.



- Client–Server, not peer-to-peer: the server holds authoritative game state (HP, mana, status effects, turn order) so both clients always see a consistent view of the match.
- Stateful / session-based: a match progresses through distinct phases (handshake → lobby/ready → combat → game over), and the server must remember where each connection is in that lifecycle.
- Turn-based, not real-time: exactly one player acts at a time; there's no strict timing requirement between messages.
- Request/response, text-based: every client message gets exactly one (or more, for broadcasts) server response, each carrying a 3-digit status code and phrase — modeled deliberately on HTTP so it's easy to log, demo, and grade.

PROGRAM CHARACTERISTIC



WHY TCP?

Reliability

If an ACTION message is lost, the game desyncs — one player thinks it's still their turn, the other is waiting forever. Turn-based state cannot tolerate silent packet loss the way, say, video streaming can.

Ordering

Status-effect ticks, action resolution, and turn-swap must happen in a fixed order. UDP does not guarantee messages arrive in the order they were sent.



WHY TCP?

Sessions

The lobby handshake (HELLO → CHOOSE → READY) naturally maps onto a persistent TCP connection per player; the server can detect disconnects (forfeits) via the socket closing.

No hard latency required

TCP's overhead is kind of irrelevant here, this is a turn-based game which takes time to think. We do not need perform every action in millisecond.

